

DoTA: Latent Distribution Conditioned Data Attribution for Diffusion Models

Ninad Joshi*, Vivek Srivastava, Shirish Karande

TCS Research, India

{ninad.j2, srivastava.vivek2, shirish.karande}@tcs.com

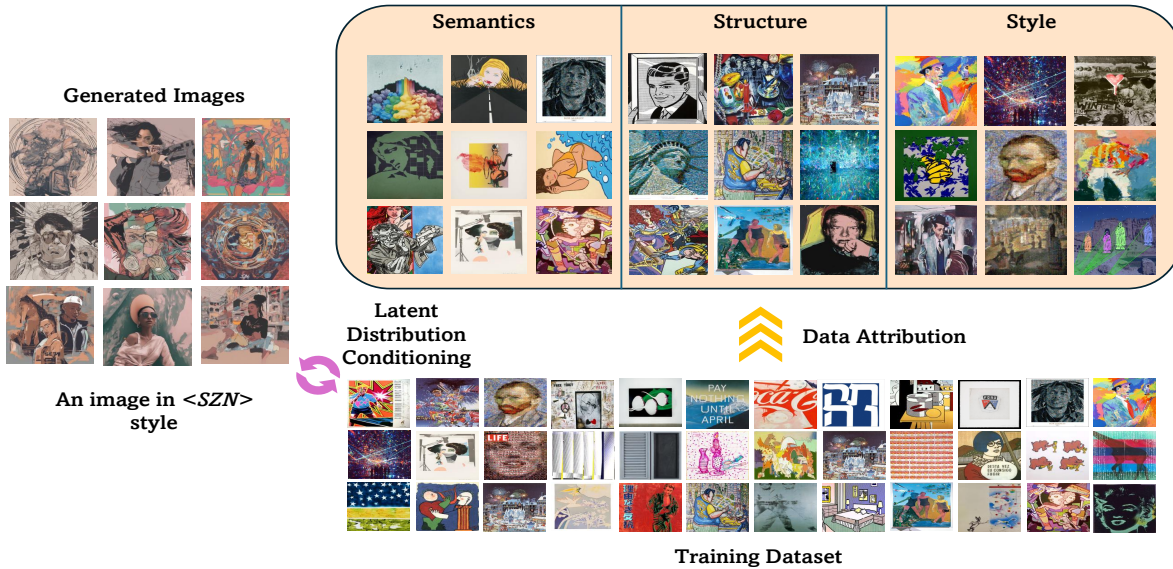


Figure 1. **Latent Distribution Conditioned Data Attribution (DoTA)**: We present a novel training data attribution method for text-to-image generation with diffusion models. Latent distribution conditioning provides fine-grained controllability for data attribution. We present three example fine-grained attribution settings: semantics, structure, and style for data attribution, where semantics refers to the object category or concept (e.g., faces), structure captures the spatial arrangement and pose, and style pertains to visual appearance aspects such as texture, color palette, or artistic rendering. The target images for data attribution are generated with the fine-tuned diffusion model that captures the style present in the images in the training dataset. The prompt used for generation includes <SZN> as a unique identifier that denotes the desired target style (e.g., PopArt), enabling consistent style control across generations.

Abstract

Diffusion models have emerged as the backbone of several modern generative AI models for effective visual content generation. However, their opaque nature raises fundamental questions about which training samples are responsible for specific generations, especially in applications involving bias detection, model auditing, and dataset curation. Data attribution seeks to identify the training samples that highly influence the output of generative models, a task that becomes especially challenging when targeting fine-scale attributes for attribution. Prior work has focused on broad concepts such as global fea-

tures or entire images, often overlooking the nuances of fine-grained attributes and relying on group-based strategies that dilute individual influence. We propose a novel latent distribution conditioned method DoTA for data attribution. DoTA presents an effective search space pruning technique based on the latent distribution matching between the generated and training data for effective and controlled attribution. We demonstrate the effectiveness of attribution through extensive quantitative and qualitative evaluations across challenging settings such as counterfactual evaluation and robustness to adversarial attack.

1. Introduction

Despite remarkable progress in generative modeling, particularly with diffusion models [12, 27, 33], a fundamental challenge remains unresolved: understanding the relationship between the training data and the generated output. This challenge, referred to as *data attribution*, is crucial in settings where accountability, interpretability, or provenance of AI-generated content is required [34, 40].

At its core, data attribution seeks to identify the subset of training examples most responsible for a given output. This is motivated by the observation that only a fraction of training samples (often called a *coreset*) significantly influence a model’s behavior. Accurately tracing this influence is essential for legal transparency, model debugging, and dataset auditing [21].

However, attributing training data to outputs becomes significantly more complex in generative models, especially diffusion-based ones [37]. Unlike classifiers, where a prediction can be directly linked to input samples, diffusion models produce images through a multi-step denoising trajectory. This makes attribution a distributed, temporal phenomenon rather than a single-step association.

Existing attribution techniques often rely on loss gradients, influence functions, or retraining heuristics. While powerful in theory, such methods are fragile in practice when applied to large, non-convex, and stochastic models. Moreover, they typically couple attribution too closely with the training algorithm, making them sensitive to initialization and hyperparameters rather than the generative behavior itself. We elaborate on these limitations in Section 2.

In this work, we propose **DoTA**, a novel framework for data attribution in diffusion models that operates at the level of latent distributional similarity. Rather than depending on gradients or retraining, our method leverages the structure of latent representations to perform search-space pruning, followed by principled attribution using Shapley values [31] over energy-based utility scores [35]. This enables interpretable, fine-grained influence estimation without sacrificing scalability.

To the best of our knowledge, ours is the first approach to jointly combine latent-level pruning with Shapley-value attribution in the context of diffusion models for text-to-image generation. We show that this not only yields faithful attributions but also leads to better robustness under adversarial and counterfactual settings.

2. Related Work

Understanding how individual training samples influence deep learning models has long been a central challenge in interpretability [10, 11, 18, 28, 39]. For generative models and diffusion models in particular, the stochasticity and non-convexity of training dynamics exacerbate this chal-

lenge, rendering many classical approaches insufficient.

A foundational line of work uses influence functions [17] to approximate the effect of upweighting a sample via loss derivatives. While theoretically elegant, their reliance on convexity and smoothness assumptions makes them unreliable in deep networks, as shown by [2]. At the other extreme, retraining-based methods [32] directly measure the impact of removing samples, offering robustness but incurring prohibitive computational costs in high-dimensional models like diffusion.

Efforts to adapt attribution to diffusion models include TRAK [26] and its extensions DTRAK [14] and JTRAK [6], which trace sample contributions across denoising steps. Although scalable, these methods depend on gradient linearity assumptions, are sensitive to stochastic sampling, and crucially require access to training logs, limiting reproducibility and general use. More efficient approaches, such as Lin et al. [20], bypass retraining via surrogate trajectory models but remain tightly coupled to internal diffusion latents, hindering applicability to unseen or closed-source architectures.

In parallel, Montrage [3] introduced a runtime attribution framework that departs from gradient or retraining-based paradigms. Instead of relying on post hoc analysis, Montrage attributes influence on-the-fly during generation by instrumenting the model’s forward computation. This design enables attribution without repeated retraining or access to training logs, offering efficiency and stability. However, it is tightly integrated with runtime execution, making it less suitable for black-box or API-based settings.

An orthogonal line leverages Shapley values [31] from cooperative game theory, offering model-agnostic guarantees of fairness and consistency. Data Shapley [9] and SHAP [23] instantiated this idea for data attribution, though their utility functions often assume retraining or simplified models. To improve scalability, KernelSHAP [23] approximated Shapley estimation via weighted regression, and RKHS-based methods [5] enabled distributional utilities. These advances allow for non-parametric objectives such as Energy Distance, which captures global distributional alignment without gradients or retraining—a property we exploit in our framework.

Other perspectives include causal approaches [13], which evaluate influence through interventions and counterfactual retraining, and machine unlearning [38], which assesses attribution by measuring performance degradation after targeted forgetting. However, both face scalability barriers due to repeated retraining or constrained optimization. Finally, similarity-based methods using pretrained encoders like CLIP [29] or DINOv2 [25] provide fast heuristics but conflate visual correlation with causal influence, limiting their faithfulness.

In summary, existing methods trade off faithfulness,

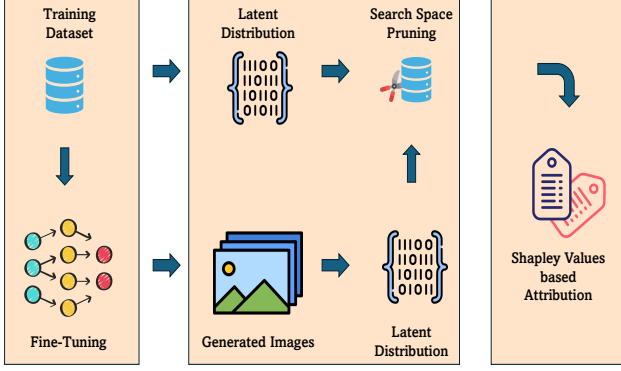


Figure 2. Architecture diagram of our proposed data attribution technique *DoTA*. We attribute the generated images on the images in the training dataset. The images are generated with a fine-tuned diffusion model on the training dataset. The latent distribution conditioned search space pruning of the training dataset allows to reduce the noise in the candidates participating in the Shapley Values-based attribution.

scalability, and architectural independence. We propose *DoTA*, a modular framework that avoids reliance on gradients, retraining, or training logs by operating directly in latent representation space and measuring statistical alignment via Energy Distance.

3. Preliminaries

3.1. Energy Distance

Energy Distance (ED) [35] is a statistical measure used to quantify the dissimilarity between two probability distributions. Unlike parametric metrics, ED is non-parametric and operates directly on samples, making it ideal for high-dimensional data such as latent representations of images. Formally, let $F_A = \{a_1, a_2, \dots, a_m\}$ and $F_B = \{b_1, b_2, \dots, b_n\}$ be two sets of latent representations drawn from distributions P and Q , respectively. The empirical Energy Distance between F_A and F_B is defined as:

$$\text{ED}(F_A, F_B) = 2 \cdot \mathbb{E}[\|a - b\|] - \mathbb{E}[\|a - a'\|] - \mathbb{E}[\|b - b'\|] \quad (1)$$

where, $a, a' \sim F_A$ are independent samples from F_A whereas $b, b' \sim F_B$ are independent samples from F_B and $\|\cdot\|$ denotes Euclidean distance. In practice, we estimate ED using finite samples from each set:

$$\begin{aligned} \text{ED}(F_A, F_B) = & \frac{2}{mn} \sum_{i=1}^m \sum_{j=1}^n \|a_i - b_j\| \\ & - \frac{1}{m^2} \sum_{i=1}^m \sum_{j=1}^m \|a_i - a_j\| \\ & - \frac{1}{n^2} \sum_{i=1}^n \sum_{j=1}^n \|b_i - b_j\| \end{aligned} \quad (2)$$

ED essentially measures how much “energy” it would take to move the points from one distribution to the other, compared to the “energy” it would take to move the points within each distribution separately.

3.2. Shapley Values

Shapley values [31], a fundamental concept from cooperative game theory introduced by Lloyd Shapley, provide a principled mechanism to fairly allocate the total value generated by a coalition among its individual members. In the context of data attribution [9], training examples are modeled as players in a cooperative game whose combined interactions contribute to the generation of an output sample. Formally, let $N = \{1, 2, \dots, n\}$ denote the set of n players (training samples), and let $v : 2^N \rightarrow \mathbb{R}$ be a utility function that assigns a real-valued score to any subset (coalition) of players. The Shapley value $\phi_i(v)$ for a player $i \in N$ is defined as:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(n - |S| - 1)!}{n!} [v(S \cup \{i\}) - v(S)] \quad (3)$$

Here:

- S is a subset of players not containing i ,
- $v(S)$ is the utility of coalition S ,
- $v(S \cup \{i\}) - v(S)$ is the marginal contribution of i ,
- The combinatorial term represents the expected contribution of i averaged over all possible permutations of the players.

We consider ED as the utility function ($v(S)$) when evaluating the closeness of a candidate subset $S \subset D$ to the generated set F_G . Specifically, we define:

$$v(S) = -\text{ED}(F_S, F_G) \quad (4)$$

where F_S denotes the projected latent representations of the training subset S . The negative sign ensures that smaller distances (greater similarity) yield higher utility scores in the Shapley framework. In our setting, the Shapley value measures the individual influence of each candidate training example on the generated image by modeling attribution as a cooperative game. This method satisfies desirable fairness properties such as efficiency, symmetry, and null player—detailed discussions of these axioms can be found in Section A.8. Appendix.

Notations Summary

- $T = \{t_1, t_2, \dots, t_n\}$: Training data latent representations, $T \in \mathbb{R}^{n \times d}$
- $G = \{g_1, g_2, \dots, g_m\}$: Generated data latent representations, $G \in \mathbb{R}^{m \times d}$
- S : A random subset of T where $|S| = p$
- δ_T : Minimum number of subset occurrences required for a training sample $t_i \in T$ to be included in the reduced candidate set T_{reduced}
- c : Minimum number of times each training sample $t_i \in T$ must be covered across randomly sampled subsets to ensure fair representation
- $\text{ED}(S, G)$: Energy Distance between subset S and the full generated set G
- δ_{ED} : q^{th} percentile of ED scores for all the training data subsets
- ϕ_i : Shapley value corresponding to the i^{th} selected training embedding
- Z : Binary matrix for SHAP sampling
- $v(S)$: Utility function (negative energy distance) for a subset S

4. Proposed Approach

In this section, we present our principled framework *DoTA* for latent distribution aware data attribution. Formally, given a generated set of images $Z_G = \{\hat{z}_1, \hat{z}_2, \dots, \hat{z}_m\}$ and a training dataset $Z_D = \{z_1, z_2, \dots, z_n\}$, our objective is to identify a subset of training examples from D that likely influenced the generation.

Setup and notations: Rather than attributing $\hat{z}$ directly based on gradient or loss-based signals, we operate in latent representation space. Specifically, we first project all images into a common latent representation space using a pretrained encoder $f(\cdot)$ followed by PCA projection for dimensionality reduction. The resulting latent representations are denoted by $T = \{t_i = f(z_i)\}_{i=1}^n$ for the training data and $G = \{g_i = f(\hat{z}_i)\}_{i=1}^m$ for the generated set.

In contrast to traditional attribution methods [6, 15, 26] that rely on gradient sensitivity or influence functions, our approach performs an early pruning of the search space based on the latent distributional similarity. This avoids exhaustive pairwise comparisons and narrows down the candidate pool to only those samples with meaningful statistical alignment.

Our framework comprises three core components. First, we **Fine-tune** the diffusion model on a small known dataset to reflect a controlled distributional shift. This step is crucial—since the original model may have been trained on unknown or proprietary data, attribution is only meaningful if we constrain the generation to examples influenced by the fine-tuning set. This makes the attribution problem well-

posed and traceable. Second, we apply a **Search Space Pruning** (SSP) strategy that leverages Energy Distance to identify training subsets closely aligned with the generated distribution. Third, we compute **Shapley Values** over the reduced candidate set to quantify the contribution of each training example. Figure 2 illustrates our proposed architecture. A detailed breakdown of each component is provided in the following subsections, along with a formal algorithmic summary (refer to Algorithm 1).

4.1. Fine-Tuning

To enable meaningful attribution, we fine-tune a pre-trained diffusion model using a known dataset. Fine-tuning acts as a controlled intervention, constraining the generative behavior to reflect the influence of a known data distribution.

4.2. Search Space Pruning

Our search space pruning approach leverages Energy Distance (ED) to isolate a subset of training examples whose latent distribution closely matches that of the generated data. By doing so, we significantly reduce the attribution search space without sacrificing fidelity.

We begin by randomly constructing subsets of training dataset $\{S_1, S_2, \dots, S_p\}$ with fixed cardinality. This is done while ensuring that every training example appears in multiple subsets to allow robust coverage across the dataset. Each subset S_i is then compared against the latent distribution of the generated images G using the ED formulation given in Equation 1. This nonparametric metric quantifies the statistical dissimilarity between the empirical distributions of the training data subset and the generated data.

After computing the ED scores for all subsets, we apply a percentile-based thresholding strategy. Specifically, we retain only those subsets whose ED scores fall below the q^{th} percentile (δ_{ED}) of the computed ED distribution, forming the passing set S_{pass} . Note that the q^{th} percentile is computed in reverse: when ED scores are sorted from low to high, we retain the top $q\%$, i.e., the subsets with the lowest divergence from the target distribution. This thresholding is designed to identify subsets with the least divergence from the target distribution, thereby ensuring that the examples within them are the most plausible contenders for actual influence. This step is crucial, as it not only reduces computational overhead but also ensures that only statistically uninformative subsets are filtered out, preserving the most relevant training examples for attribution. (refer to Algorithms 2 and 3)

Next, we aggregate all training examples that appear in at least δ_T passing subsets, yielding a reduced candidate set $T_{\text{reduced}} \subset T$. These examples are considered statistically consistent with the generated data and are deemed most likely to have contributed to the generation of $\hat{z}$. This reduced pool of candidates is then used for Shapley values-

Algorithm 1 DoTA

Require: Training embeddings T , Generated embeddings G

- 1: $S_{\text{pass}} \leftarrow \text{Algorithm 2}$ (Data Subset Selection)
 - 2: $T_{\text{reduced}} \leftarrow \text{Algorithm 3}$ (Search Space Pruning)
 - 3: **for** each $g_i \in G$ **do**
 - 4: $\phi^{(i)} \leftarrow \text{Algorithm 4}$ (KernelSHAP on T_{reduced} and g_i)
 - 5: **end for**
 - 6: **return** $\{\phi^{(i)}\}_{i=1}^{|G|}, T_{\text{reduced}}$
-

Algorithm 2 Data Subset Selection

Require: Training set T , generated set G , subset size p , minimum coverage c , percentile q

- 1: Initialize list of random subsets $\mathcal{S} = \{\}$
 - 2: Initialize coverage counter for each $t_i \in T$
 - 3: **while** minimum coverage of any t_i is $< c$ **do**
 - 4: Sample a random subset $S \subset T$ such that $|S| = p$
 - 5: Add S to $\mathcal{S}$, increment counters of elements in S
 - 6: **end while**
 - 7: Compute $\text{ED}(S_i, G)$ for each $S_i \in \mathcal{S}$
 - 8: Compute $\delta_{\text{ED}} = \text{Percentile}(\{\text{ED}(S_i, G)\}_i, q)$
 - 9: $S_{\text{pass}} = \{S_i \in \mathcal{S} \mid \text{ED}(S_i, G) \leq \text{Threshold}\}$
 - 10: **return** S_{pass}
-

Algorithm 3 Search Space Pruning

Require: Subset list S_{pass} , threshold δ_T (e.g., 2)

- 1: Initialize count map $C[t_i] \leftarrow 0$ for all $t_i \in T$
 - 2: **for** each $S \in S_{\text{pass}}$ **do**
 - 3: **for** each $t_i \in S$ **do**
 - 4: $C[t_i] \leftarrow C[t_i] + 1$
 - 5: **end for**
 - 6: **end for**
 - 7: $T_{\text{reduced}} = \{t_i \in T \mid C[t_i] \geq \delta_T\}$
 - 8: **return** T_{reduced}
-

based attribution.

Note that, directly comparing the full training set with a small generated set via ED can be unstable due to unequal sample sizes, which skews the pairwise distance terms. To avoid this, we compare equally sized random subsets of the training data with the generated set, enabling fairer and more stable ED estimates. This also allows us to model higher-order interactions and retain only those samples that consistently align well across multiple low-ED subsets, making the reduced candidate set both statistically robust and computationally tractable.

4.3. Shapley Value-Based Attribution

After pruning the candidate set T_{reduced} , we estimate the individual influence of each training example on the generated image $\hat{z}$ using Shapley values. We treat each candidate as a player in a cooperative game, where the utility of a subset $S \subseteq T_{\text{reduced}}$ is defined by the negative ED between S and the generated set G as given in Equation 4. To efficiently

Algorithm 4 KernelSHAP on Pruned Training Set

Require: Reduced training set T_{reduced} , generated image g , number of samples M

- 1: Let $d \leftarrow |T_{\text{reduced}}|$
 - 2: Estimate mean energy $\bar{E}$ using random subsets of size p
 - 3: Initialize dataset $\mathcal{D} = \{\}$
 - 4: **for** $j = 1$ to M **do**
 - 5: Sample binary mask $z \in \{0, 1\}^d$ with exactly p ones
 - 6: Let $S = \{t_i \in T_{\text{reduced}} \mid z_i = 1\}$
 - 7: Compute utility: $v(S) = -(\text{ED}(S, \{g\}) - \bar{E}) \cdot \text{scale}$
 - 8: Compute kernel weight $w(z)$ using SHAP kernel
 - 9: Append $(z, v(S), w(z))$ to $\mathcal{D}$
 - 10: **end for**
 - 11: Fit weighted linear regression: $v(S) \approx z^\top \phi$ using weights $w(z)$
 - 12: **return** Shapley values $\phi \in \mathbb{R}^d$
-

approximate the Shapley values, we adopt a KernelSHAP [23] based linear regression. We randomly sample M subsets with binary masks and regress their corresponding utilities using a weighted model:

$$\phi = \arg \min_{\phi} \sum_{m=1}^M w^{(m)} \left(v(S_m) - \langle z^{(m)}, \phi \rangle \right)^2, \quad (5)$$

where $w^{(m)}$ is the standard SHAP kernel weight, and ϕ_i gives the attribution score for image $x_i \in T_{\text{reduced}}$. This produces a Shapley value vector ϕ capturing the relative influence of each candidate on $\hat{z}$ (refer to Algorithm 4).

The inclusion of the scaling factor in the utility computation serves to amplify the dynamic range of the values provided to the weighted regression estimator. Since KernelSHAP estimates Shapley values by solving a weighted linear regression problem, very small or tightly clustered utility scores can lead to ill-conditioned regression matrices and numerical instability, especially when ridge regularization is applied. By scaling the utilities, we effectively increase variance in the response variable, improving the conditioning of the regression problem and enabling more reliable estimation of attribution weights. Importantly, the scaling does not alter the relative ordering of coalition utilities, preserving the consistency of the underlying Shapley formulation while enhancing estimation accuracy.

5. Experiments

5.1. Dataset and Fine-tuning

Dataset: We evaluate our approach on three diverse art collections: (i) PopArt (1,482 artworks from WikiArt [36]), (ii) Ukiyo-e (2,000 prints from the Metropolitan Museum of Art’s digital collection [16]), and (iii) Edvard Munch Archive (5891 works from the digital archive [24]). These datasets vary in scale and capture unique fine-grained artistic details, providing a robust testbed for attribution. Addi-

Method	Relative Style Confidence					
	Semantics		Structure		Style	
	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)
Cosine Similarity	0.0320	0.0275	0.0324	0.0399	0.0313	0.0290
DoTA wo SSP	0.0523	0.0483	0.0432	0.0335	0.0498	0.0294
DoTA with SSP	0.0370	0.0238	0.0484	0.0327	0.0374	0.0361
Absolute Style Confidence						
Cosine Similarity	0.3463	0.3406	0.3628	0.3553	0.3313	0.3565
DoTA wo SSP	0.4047	0.4010	0.3485	0.3561	0.4219	0.3799
DoTA with SSP	0.3667	0.3648	0.3784	0.3300	0.3784	0.3783

(a) PopArt

Method	Relative Style Confidence					
	Semantics		Structure		Style	
	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)
Cosine Similarity	0.0313	0.0424	0.0227	0.0139	0.0303	0.0313
DoTA wo SSP	0.0323	0.0204	0.0348	0.0268	0.0372	0.0332
DoTA with SSP	0.0466	0.0303	0.0445	0.0245	0.0479	0.0394
Absolute Style Confidence						
Cosine Similarity	0.3568	0.3690	0.4121	0.3707	0.3273	0.3241
DoTA wo SSP	0.4752	0.3234	0.4439	0.3543	0.5278	0.3373
DoTA with SSP	0.4510	0.3318	0.4927	0.3674	0.5641	0.3534

(b) Ukiyo-e

Method	Relative Style Confidence					
	Semantics		Structure		Style	
	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)
Cosine Similarity	0.0279	0.0215	0.0137	0.0372	0.0122	0.0207
DoTA wo SSP	0.0150	0.0168	0.0334	0.0286	0.0287	0.0194
DoTA with SSP	0.0314	0.0489	0.0209	0.0206	0.0298	0.0216
Absolute Style Confidence						
Cosine Similarity	0.2851	0.2988	0.2679	0.3129	0.2511	0.2669
DoTA wo SSP	0.2905	0.2896	0.2947	0.3044	0.3154	0.2896
DoTA with SSP	0.3106	0.3291	0.2853	0.3173	0.3179	0.2848

(c) Munch

Table 1. **Counterfactual evaluation** of attribution methods for the three datasets with representation dimensions reduced to 50. We highlight the best result for each counterfactual setting for a latent representation with blue color. The cases where the style confidence for Leave-K is higher than Take-K (where K = 5%) are highlighted with red color.

tional details and examples are provided in the Appendix.

Fine Tuning: We perform LoRA-based fine-tuning of the Stable Diffusion XL model (SDXL [27]). We adapt the fine-tuning strategy presented in DreamBooth [30] which enables efficient adaptation of both object and style representations through low-rank updates. The model is trained using the AdamW [22] optimizer with a weight decay of 1e-04 for Unet and 1e-03 for text encoder, for 1000 training steps. The prompt is set to “an image in <SZN> style” for each image where the identifier <SZN> captures the details from the training data. During inference, the same keyword is used to recreate the learned details from the training data (refer to Section A.3. and B.1. in Appendix).

5.2. Fine-Scale Attribute Representations

To evaluate the training-to-generation alignment across fine-granular visual dimensions, we use three fine-scale latent representations tailored to capture unique visual attribute. We specifically adopt CLIP [29] for semantic details, DINOv2 [25] for structural details, and Gram Matrix [8] features for stylistic details, following prior works

Method	Relative Style Confidence					
	Semantics		Structure		Style	
	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)
Cosine Similarity	0.0672	0.0620	0.0655	0.0597	0.0743	0.0769
DoTA wo SSP	0.0685	0.0631	0.0586	0.0662	0.0608	0.0600
DoTA with SSP	0.0640	0.0553	0.0663	0.0479	0.0589	0.0589
Absolute Style Confidence						
Cosine Similarity	0.4939	0.5007	0.5324	0.5028	0.5122	0.5285
DoTA wo SSP	0.4741	0.5144	0.4905	0.5181	0.4962	0.4962
DoTA with SSP	0.5220	0.4921	0.4978	0.4796	0.5274	0.4956

(a) PopArt

Method	Relative Style Confidence					
	Semantics		Structure		Style	
	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)
Cosine Similarity	0.0414	0.0414	0.0391	0.0458	0.0482	0.0514
DoTA wo SSP	0.0398	0.0446	0.0435	0.0394	0.0396	0.0380
DoTA with SSP	0.0427	0.0328	0.0464	0.0352	0.0523	0.0317
Absolute Style Confidence						
Cosine Similarity	0.4786	0.4740	0.4686	0.4721	0.4621	0.4846
DoTA wo SSP	0.4662	0.4706	0.4843	0.4727	0.4586	0.4523
DoTA with SSP	0.4967	0.4420	0.4721	0.4451	0.4840	0.4491

(b) Ukiyo-e

Method	Relative Style Confidence					
	Semantics		Structure		Style	
	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)	Take-K (↑)	Leave-K (↓)
Cosine Similarity	0.0955	0.0886	0.1211	0.1085	0.0942	0.0801
DoTA wo SSP	0.0909	0.1124	0.0976	0.0834	0.0992	0.0693
DoTA with SSP	0.1187	0.0953	0.0928	0.0900	0.0813	0.0846
Absolute Style Confidence						
Cosine Similarity	0.5054	0.4971	0.5182	0.5094	0.5118	0.4819
DoTA wo SSP	0.4975	0.4929	0.5294	0.4945	0.5059	0.4997
DoTA with SSP	0.5169	0.5543	0.5221	0.4941	0.4739	0.4735

(c) Munch

Table 2. **Counterfactual evaluation under adversarial attack** of attribution methods for the three datasets with representation dimensions reduced to 50. We highlight the best result for each counterfactual setting for a latent representation with blue color. The cases where the style confidence for Leave-K is higher than Take-K (where K = 5%) are highlighted with red color.

that highlight the suitability of each representation for these roles [1, 7, 19] (refer to Section A.4. in Appendix).

- **Semantics.** Semantic similarity is captured using CLIP embeddings [29], which are known to align images and text in a shared space and are widely adopted to assess high-level conceptual attributes such as object identity, scene context, and category-level meaning [1, 7].
- **Structure.** For structural content, we use DINOv2 embeddings [25], a self-supervised vision transformer trained with a distillation-based objective. DINOv2 preserves mid-level features like object boundaries, shape configurations, and spatial layout—key for understanding structural alignment [4].
- **Style.** Stylistic aspects are modeled using Gram matrices of deep convolutional features from a VGG-19 network [8]. These matrices capture second-order statistics between feature maps, reflecting texture, color repetition, and brushstroke patterns, and have been extensively used in style transfer literature [19].

5.3. Evaluation Setup

In this Section, we present our evaluation setup for the data attribution. As a baseline, to quantitatively measure the alignment between a generated image and candidate training samples, we employ cosine similarity in the latent space. This choice is intentional: unlike methods such as TRAK [26] and its generative variants (DTRAK, JTRAK), which rely on approximating gradients or loss surface behavior, cosine similarity provides a stable, non-parametric, and model-agnostic measure of angular closeness between latent representations. Influence function-based methods [18] are known to break down in deep, non-convex models due to high variance and fragility in stochastic regimes [2]. Similarly, unlearning-based approaches [38] require costly retraining or auxiliary constraints like the Fisher Information Matrix to avoid catastrophic forgetting, making them less scalable and unsuitable for fine-grained attribution. While runtime attribution frameworks such as Montrage [3] avoid retraining and gradient reliance by instrumenting the generation process directly, they require modifications to the internal forward execution of the model and continuous access to training logs. In contrast, cosine similarity is computationally efficient, robust across encoder choices, and captures structural alignment in the representation space—making it better suited for our attribute-aware evaluation framework.

5.3.1. Counterfactual Evaluation

Here, we evaluate how removing or exclusively using top-ranked influential examples alters the model’s ability to reproduce the targeted generative behavior. We use two complementary setups:

1. **Removal-Based Evaluation (Leave-K):** We retrain the generative model after removing the top- K most positively influential training samples. We then measure the relative change in the model’s ability to regenerate the target images.
2. **Reconstruction-Based Evaluation (Take-K):** In this setting, we train a model *only* on the top- K most influential examples and assess whether it alone can reconstruct the reference behavior.

In our experiments, we choose K as the top 5% samples from the attributed images. To quantify the change in model behavior, we employ a **multi-class style classifier** trained on multiple artistic styles, as proposed in the UnlearnCanvas framework [41]. This classifier allows us to assess whether counterfactual generations maintain alignment with the intended target style. We consider both the *classification probability* assigned to the target style and its *rank* among all style classes. We normalize the target class probability by its rank, ensuring that both absolute confidence and relative priority are captured. Specifically, we denote the average classification probability of the

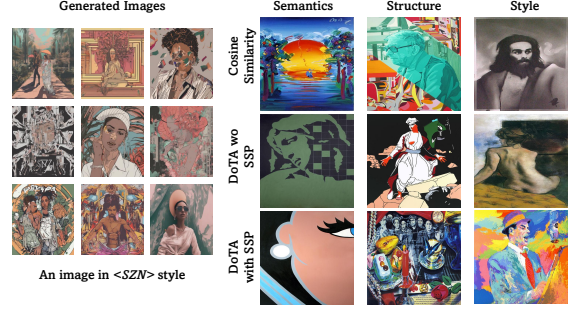


Figure 3. Qualitative comparison of different approaches for data attribution on the PopArt dataset. We present the **best attributed image** with each method across representations. The best attributed images with DoTA effectively capture the semantics, structure, and style present in the generated images.

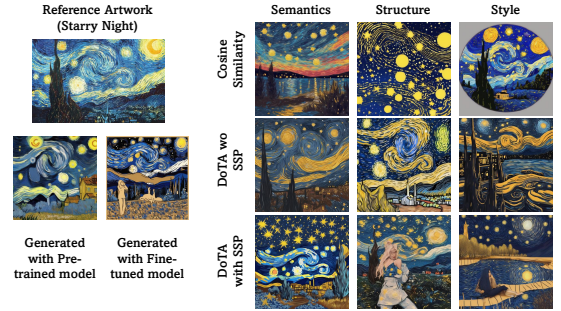


Figure 4. Qualitative comparison of different approaches under adversarial attack with different latent representations on the PopArt dataset. We present the **images generated for “Starry Night” after retraining with Take-K under adversarial attack** in different settings. We observe maximum impact of the attack with the cosine similarity based attribution showing drastic changes in the generated images.

target style as **Absolute Style Confidence (ASC)**, and the rank-normalized probability as **Relative Style Confidence (RSC)**. An effective attribution method should simultaneously increase the target style confidence in the Take-K setting and reduce it in the Leave-K setting. In both settings, we generate target images with the prompt “an image in $\langle \text{SZN} \rangle$ style” and report the ASC and RSC metric scores.

5.3.2. Certifiability under Adversarial Attack

We further evaluate the quality of the attribution under adversarial attack. We introduce an adversarial set of 25 popular artworks from five different artists (refer to Section B.4. in Appendix). This adversarial set is only used during attribution and not during model fine-tuning. The model fine-tuning is performed only on the images in the training dataset. We consider both counterfactual settings (Leave-K and Take-K) for evaluation. To generate target images during attribution, we change the prompts to “ $\langle \text{popular} \rangle$

artwork> in <SZN> style” where we replace <popular artwork> with the name of the artwork. While retraining the model for Leave-K and Take-K experiments, we exclude samples from the adversarial set and only use attributed images from the dataset. A robust attribution method should focus on training-set samples, ensuring that counterfactual evaluations are minimally impacted by adversarial attacks.

5.4. Results and Analysis

We present the counterfactual evaluation for style fidelity in Tables 1 and 2. We observe that DoTA consistently shows superior performance compared to cosine similarity-based attribution across representations and both counterfactual settings. However, in some cases, we observe an interesting behavior where the style confidence for Leave-K is higher than Take-K. For cosine similarity, we attribute this behavior to its inability to effectively capture attribute-conditioned similarity in the latent distribution during attribution. In contrast, DoTA without SSP considers the entire training dataset for Shapley value-based attribution, which often introduces noise during coalition formation. DoTA with SSP is largely free from this behavior, except for a few instances on the Munch dataset, whereas cosine similarity-based attribution consistently struggles due to the extremely high diversity in the dataset. Furthermore, we observe a significant reduction in computation time with DoTA when incorporating the SSP technique (refer to Table 3 and Section B.5. in the Appendix). However, the memory usage is higher for SSP compared to its non-SSP counterpart. A plausible reason is that SSP relies on a larger number of simulations during Algorithm 2 (Data Subset Selection) and Algorithm 3 (Search Space Pruning), which require storing the Energy Distance (ED) values for multiple random subsets $\mathcal{S}$. These mappings between subsets and their corresponding ED values are subsequently reused in Algorithm 4 for ridge regression to compute the Shapley weights, thereby incurring additional memory overhead.

We observe a similar pattern in the counterfactual evaluation under adversarial attack, where both cosine similarity and DoTA without SSP show limited robustness. In contrast, our proposed SSP-enhanced Shapley value-based attribution demonstrates significant improvement by reducing noise in the candidate set participating in attribution. We present qualitative results in Figure 3 and discuss them in detail in Section B.5. of the Appendix. The best-attributed images with DoTA effectively capture the semantics, structure, and style of the generated images. For instance, the majority of the generated images depict a woman, which is faithfully reflected in the semantics-aware attributions obtained with DoTA.

The results in Table 4 quantify how well attribution methods can resist the influence of adversarial intrusions of popular artworks injected into the training set. DoTA-based

Method	PopArt					
	Semantics		Structure		Style	
	Time	Memory	Time	Memory	Time	Memory
Cosine	4763.84	1189.45	5429.43	1291.87	7542.74	1558.83
DoTA wo SSP	1592.12	324.58	1624.71	517.42	1973.91	735.41
DoTA with SSP	11.73	549.87	14.33	820.31	35.41	1071.87
Method	Ukiyo-e					
	Semantics		Structure		Style	
	Time	Memory	Time	Memory	Time	Memory
Cosine	10122.66	1427.83	12728.32	1768.72	14327.22	2136.42
DoTA wo SSP	1918.36	595.79	1925.78	594.81	1900.81	786.32
DoTA with SSP	12.09	779.11	13.94	780.02	451.04	1093.56
Method	Munch					
	Semantics		Structure		Style	
	Time	Memory	Time	Memory	Time	Memory
Cosine	35871.21	5874.73	37287.92	6174.87	45879.12	6843.41
DoTA wo SSP	19164.86	769.45	20831.71	1377.36	21536.87	1591.11
DoTA with SSP	894.07	944.32	978.87	1527.32	1137.28	1879.23

Table 3. **Attribution time and run time memory allocation.** We report the results for attribution performed with representation dimensions reduced to 50. We report the time in seconds and memory in megabytes.

Method	PopArt			Ukiyo-e			Munch		
	Sem	Str	Sty	Sem	Str	Sty	Sem	Str	Sty
Cosine Similarity	88	92	68	60	48	72	56	48	72
DoTA wo SSP	56	32	8	40	56	64	32	36	48
DoTA with SSP	48	40	16	20	28	40	20	28	40

Table 4. Percentage of samples from the adversarial examples attributed in the Top-K. Lower values indicate better certifiability under adversarial attack. We highlight the best and worst scores with blue and red colors respectively.

attribution demonstrates robust performance against adversarial attacks across all latent representations. Furthermore, we present qualitative results in Figure 4 and “The Scream” in Appendix. The images generated for “Starry Night” after retraining with Take-K under adversarial attack show a significant loss of semantics, structure, and style for cosine similarity-based attribution, whereas DoTA shows minimal changes in the generated images across all three latent representations.

6. Conclusion

In this work, we introduced a latent distribution conditioned attribution framework for diffusion models that combines energy-distance-based pruning with cooperative game theory to enable effective and robust attribution. Our results under adversarial settings highlight its resilience, while pointing toward future directions in efficient calibration for faster attribution. We believe these advances lay the groundwork for practical applications in certifiability of visual art and agency attribution in co-creative systems.

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